

Reinforcement Learning for Economically Optimized Churn Management: A Contextual Bandit Framework

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Abstract—This study presents a novel application of reinforcement learning for customer churn management using a Contextual Bandit (CB) approach. Building upon the limitations of static machine learning classifiers, we implemented a Thompson Sampling Bandit agent that integrates churn probabilities, Customer Lifetime Value (CLV), and intervention costs into a real-time decision-making framework. Unlike traditional models focusing solely on prediction accuracy, our approach optimizes promotional targeting to maximize economic return. The results demonstrate that the contextual bandit learns optimal promotional policies and adapts to segment-specific behaviors, outperforming rule-based methods in economic utility. The paper illustrates the feasibility and effectiveness of reinforcement learning in retention-focused CRM systems.

Keywords—Customer churn prediction, reinforcement learning, Contextual Bandits, Thompson Sampling, Customer Lifetime Value (CLV), economic optimization, personalized retention.

I. INTRODUCTION

Customer churn remains a primary challenge for businesses operating in competitive and subscription-driven markets. With the cost of acquiring new customers significantly exceeding the cost of retaining existing ones, effective churn prediction and targeted retention strategies have become essential for ensuring long-term profitability and customer base stability. Traditional predictive modeling methods such as decision trees, logistic regression, and neural networks have been widely employed to forecast the likelihood of customer attrition. However, these models often prioritize classification accuracy without explicitly accounting for the economic consequences of misclassification or intervention decisions.

Scientific research has highlighted the importance of incorporating economic rationality and behavioral insights into churn management. In their foundational work, Kahneman and Tversky [1] introduced the Prospect Theory, emphasizing that human uncertainty-based decision-making is influenced by perceived gains and losses rather than expected utility. This perspective is crucial for churn modeling, where businesses must decide whether to invest in potentially costly retention efforts based on probabilistic predictions of customer behavior.

Building upon these ideas, Verbraken et al. [2] proposed a profit-maximizing metric for evaluating churn prediction models, advocating for a performance assessment framework grounded in financial return rather than predictive accuracy alone. Similarly, Bahnsen et al. [3] developed a cost-sensitive classification framework that integrates customer value and

intervention costs into the model training process, enhancing the alignment between model outputs and actionable business strategies.

Further, recent advances in group decision-making and consensus modeling, such as the work by Yuan et al. [4], emphasize the importance of adapting decision strategies to heterogeneous and potentially conflicting behaviors within customer populations. These approaches stress the need for adaptive, personalized, and behavior-aware models in domains characterized by dynamic and uncertain environments.

Despite these advances, existing models often operate in static settings and lack the capacity to learn and adapt in real time based on observed outcomes. To address this gap, we propose a reinforcement learning approach—specifically, a Thompson Sampling-based Contextual Bandit framework—that dynamically optimizes retention decisions by balancing exploration of uncertain actions and exploitation of known profitable strategies. The proposed model advances beyond traditional predictive analytics toward economically optimized and context-aware customer relationship management by incorporating churn probabilities, customer lifetime value (CLV), and intervention costs into a sequential decision-making process.

II. METHODOLOGY

The proposed methodology integrates supervised churn prediction with reinforcement learning in the form of contextual bandits, guided by principles from economic utility theory. By leveraging both predictive modeling and adaptive decision-making, the framework supports intelligent, profit-aware customer retention strategies that adjust to individual behavior and business constraints.

A. Data Origin and Ethical Considerations

The dataset used in this study was obtained from a publicly available Kaggle repository, consisting of anonymized transactional and behavioral records from an online retail platform. Although no personally identifiable information is present, the dataset realistically reflects real-world patterns of customer engagement, churn behavior, and economic characteristics such as purchase frequency and Customer Lifetime Value (CLV). The use of this dataset enables reproducibility while complying with ethical data usage standards. All pre-processing and feature engineering steps were conducted in line with best practices for responsible data science and customer analytics.

B. Exploratory Data Analysis (EDA)

Initial exploratory data analysis was conducted to understand customer behavior patterns, identify feature distributions, and detect data quality issues. Key findings included:

- Churn distribution - approximately 55% of the customers were labeled as churned, indicating a moderately imbalanced dataset.
- CLV variability - CLV values exhibited significant skew, with a subset of high-value customers contributing disproportionately to total revenue.
- Feature correlation - a high correlation was observed between churn and variables such as return ratio, low engagement score, and short tenure.
- Promotion responsiveness - customers with high engagement scores showed higher promotion response rates but did not always translate into lower churn risk.

This analysis informed the subsequent steps in feature engineering, modeling, and reward design..

C. Data Processing and Engineering

The raw dataset included demographic, transactional, and engagement variables for 600 customers. Preprocessing steps included:

- Data Cleaning: Imputation of missing values and removal of outliers based on Z-score thresholds.
- Encoding: Categorical features (e.g., gender, promotional status) were transformed using one-hot encoding.
- Scaling: Numerical variables were standardized to zero mean and unit variance to ensure model stability.

Derived features were generated to better capture behavioral trends:

- Return Ratio: $\text{Return Ratio} = \frac{\text{Number of Returns}}{\text{Total Purchases}}$
- Purchase Frequency:

$$\text{Purchase Frequency} = \frac{\text{Purchases}}{\text{Days Since First Purchase}}$$

- Engagement Score: A weighted index combining email open rates, promotional clicks, and customer service interactions.
- Customer Lifetime Value (CLV) estimated using:

$$\text{CLV} = \text{Average Purchase Value} \times \text{Purchase Frequency} \times \text{Customer Tenure}$$

These variables enhanced model accuracy and contributed to a financially grounded reward structure in the reinforcement learning component.

D. Customer Utility Function Definition

Inspired by Prospect Theory [1], we defined a customer utility function to encode perceived value and risk aversion:

$$U(\text{Customer}) = w^+(P_{\text{stay}}) \times V(\text{Gain}) - w^-(P_{\text{churn}}) \times V(\text{Loss})$$

Where:

- $w^+(\cdot)$ and $w^-(\cdot)$ are probability weighting functions modeling behavioral biases;

- $P_{\text{stay}} = 1 - P_{\text{churn}}$ is derived from the supervised model;

- $V(\cdot)$ is the subjective valuation of financial outcomes, anchored by CLV and intervention costs.

In our reinforcement learning implementation, this behavioral model was approximated by a reward function (see subsection E), directly incorporating CLV and intervention cost as proxies for perceived gain and loss. The simplified utility function ensures tractability within the Thompson Sampling framework while retaining economic relevance. Specifically, the expected utility, — calculated as $E[U] = P(\text{churn}) \cdot \text{CLV} - \text{Intervention Cost}$ — was used as a reward signal guiding the agent's decisions. During each interaction, this value was passed to the contextual bandit, enabling it to adaptively prioritize high-value customer actions and optimize retention outcomes over time.

E. Contextual Bandit Framework

The agent operates under the *Contextual Bandit* formulation, wherein each decision is made based on a context vector $x \in \mathbb{R}^d$ associated with a customer.

- *Action space* $A = \{0,1\}$:

$a = 0$: no action taken;

$a = 1$: promotion sent.

- *Reward* r is determined by the customer's simulated response (see subsection E) and economic impact.

This framework facilitates customer-specific intervention decisions while supporting real-time updates and scalability.

F. Thompson Sampling for Policy Learning

To solve the contextual bandit problem, we employ the *Thompson Sampling* algorithm, which maintains posterior Beta distributions $\text{Beta}(\alpha_a, \beta_a)$ over the expected rewards of each action a . The algorithm proceeds as follows:

- 1) Sampling: For each action, draw a sample $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$
- 2) Action selection: Choose $a^* = \arg \max_a \theta_a$
- 3) Reward observation: Based on churn simulation and CLV-based reward.
- 4) Bayesian update: Update α_a and β_a using reward > 0 or ≤ 0 .

The Thompson Sampling contextual bandit was implemented using a Bayesian linear regression prior over the neural representation of the customer context. The model was trained over 1,000 simulated episodes, with exploration controlled through posterior sampling on each decision round. Hyperparameters, including prior variance and learning rate, were optimized via grid search using average cumulative utility as the objective criterion.

This approach ensures an optimal balance between *exploration* (testing new policies) and *exploitation* (maximizing known returns).

G. Economic Reward Modeling

The design of the reward function is central to aligning agent behavior with business objectives. Reward values were clipped to a bounded range $[-1, 1]$ to stabilize learning and

avoid outlier bias from extremely high CLV values. In cases where no intervention was applied, the baseline utility was set to zero, serving as a control against which intervention outcomes were compared. This normalization ensured the agent’s learning remained focused on relative utility improvements rather than absolute values. Let c be the fixed cost of a promotion, and V the customer’s estimated CLV.

The reward structure is defined as:

$$r = \begin{cases} V - c, & \text{if action} = 1 \text{ and customer stays} \\ -c, & \text{if action} = 1 \text{ and customer churns} \\ V, & \text{if action} = 0 \text{ and customer stays} \\ 0, & \text{if action} = 0 \text{ and customer churns} \end{cases}$$

This formulation captures:

- the opportunity cost of taking no action;
- the risk of unnecessary promotions;
- the value of successfully retained customers.

These reward normalization techniques helped maintain the interpretability of financial outcomes when comparing agent-based decisions to static or random policies. Such a reward design transforms the classical classification problem into a *utility-driven control problem*.

H. Agent Integration in Decision Pipeline

The contextual bandit agent is embedded within a pipeline that processes customer records sequentially. For each customer:

- 1) Compute feature vector x .
- 2) Predict churn probability using a supervised model.
- 3) Construct context and apply Thompson Sampling to select action.
- 4) Log decisions and rewards for evaluation and audit.

Additionally, segmentation layers (e.g., CLV quartiles and churn risk bands) allow for aggregated policy analysis and budget optimization. This methodology facilitates an integrated approach to churn management, where predictive analytics, behavioral economics, and sequential decision-making are synthesized into a single, interpretable, and adaptive system.

III. RESULTS

To evaluate the effectiveness of the proposed reinforcement learning framework, a series of simulations were conducted using real-world customer data and model-generated churn probabilities. The results presented in this section illustrate the agent’s learning behavior, action selection trends, and the economic impact of its decisions across different customer segments. Particular attention is given to the distribution of rewards, convergence dynamics, and the agent’s ability to differentiate between profitable and unprofitable retention opportunities. The model was implemented in Python using Pandas, NumPy, Scikit-learn, and Matplotlib.

The initial stage of the analysis focused on understanding the customer dataset through exploratory techniques. Key behavioral indicators—such as product return ratio, purchase frequency, and promotional responsiveness—were found to exhibit meaningful variation across churned and retained customers. A simplified Customer Lifetime Value (CLV)

formula combined income-based potential with historical spending and loyalty-adjusted tenure. A correlation analysis evaluated the linear relationship between customer attributes and the churn indicator. As shown in Table I, none of the individual features exhibited a strong correlation with churn, with all correlation coefficients falling below ± 0.05 .

The variables with the highest absolute correlations were *Annual_Income* (-0.036), *Num_of_Purchases* (-0.033), and *Years_as_Customer* (-0.030), all showing weak negative associations with churn. Interestingly, *CLV* and *Total_Spend*—despite their financial significance—demonstrated near-zero correlations (0.0026 and 0.0287 , respectively), suggesting that raw monetary metrics alone may not sufficiently explain customer attrition. These findings reinforce the importance of using multivariate modeling and utility-driven formulations to capture complex churn dynamics rather than relying on univariate relationships.

In the second phase of the research, a utility-based formulation of customer behavior was developed using principles from Cumulative Prospect Theory (CPT). The goal was to express each customer’s perceived value of promotional interventions, incorporating psychological factors such as loss aversion and nonlinear probability weighting.

To operationalize the CPT framework, utility values were computed for each customer using normalized satisfaction scores as proxies for perceived gain and loss. Table II presents the first 10 computed records, where the transformed satisfaction score informs two outcome-specific value functions: V_{Gain} and V_{Loss} . Customers with high satisfaction scores (e.g., Customers 4 and 5) exhibit zero perceived loss and maximal gain, resulting in high overall utility. Conversely, customers with low satisfaction (e.g., Customer 7) showed a high perceived loss, producing significantly negative utility values.

TABLE I. CORRELATION WITH CHURN

Feature	Correlation
Target_Churn	1.000000
Target_Churn_Binary	1.000000
Customer_ID	0.043034
Total_Spend	0.028659
Satisfaction_Score	0.022567
Num_of_Support_Contacts	0.009593
CLV	0.002607
Num_of>Returns	0.000061
Email_Opt_In	-0.001019
Age	-0.009260
Last_Purchase_Days_Ago	-0.013823
Average_Transaction_Amount	-0.024723
Years_as_Customer	-0.029823
Num_of_Purchases	-0.032772

TABLE II. CUSTOMER UTILITY FUNCTION BASED ON CPT (FIRST 10 ROWS)

Customer ID	Satisfaction Score	Normalized Satisfaction	V_{Gain}	V_{Loss}	$w^+(P_{stay})$	$w^-(P_{churn})$	Customer Utility
1	3	0.6	0.637930	1.004605	0.433141	0.466053	-0.191885
2	3	0.6	0.637930	1.004605	0.433141	0.466053	-0.191885
3	2	0.4	0.446491	1.435343	0.433141	0.466053	-0.475552
4	5	1.0	1.000000	0.000000	0.433141	0.466053	0.433141
5	5	1.0	1.000000	0.000000	0.433141	0.466053	0.433141
6	4	0.8	0.821711	0.545870	0.433141	0.466053	0.101513
7	1	0.2	0.242609	1.848850	0.433141	0.466053	-0.756579
8	3	0.6	0.637930	1.004605	0.433141	0.466053	-0.191885
9	4	0.8	0.821711	0.545870	0.433141	0.466053	0.101513
10	2	0.4	0.446491	1.435343	0.433141	0.466053	-0.475552

Despite uniform probability weights across the segment (due to constant churn probability input in this stage), the variation in utility stemmed from the behavioral nonlinearity of value perception. The *Customer_Utility* column, aggregating weighted gain and loss valuations, reflects the subjective desirability of promotional engagement. These computed utilities were later used as economic signals within the reinforcement learning framework to optimize intervention strategies based on profit-driven behavioral insights.

In the third stage, the dataset was transformed to support machine learning modeling and utility-based reinforcement learning. Several key preprocessing and feature engineering steps were implemented:

- variables such as gender, promotional response, and email opt-in status were transformed via one-hot encoding to ensure compatibility with supervised models;
- *Return_Ratio* and *Purchase_Frequency* quantify post-purchase behavior and transaction intensity, respectively, capturing dissatisfaction and recency-driven loyalty;
- *Engagement_Score* reflects the degree of responsiveness to marketing communication, ranging from 0 (unengaged) to 4 (highly engaged);
- *Customer Lifetime Value (CLV)* was re-estimated using enriched transactional data, ensuring consistent alignment with previously calculated utility metrics. It serves as a core economic indicator, with values spanning from below 100 to over 5,000, emphasizing the heterogeneity in customer profitability;
- a uniform *Intervention_Cost* of 50 was introduced to simulate the expense associated with executing a promotional retention strategy.

This prepared dataset formed the input for predictive churn modeling, enabling subsequent steps to incorporate both behavioral signals and economic prioritization in the design of adaptive retention strategies.

The next research phase introduced a contextual bandit framework to enable adaptive decision-making in promotional targeting. The bandit agent was trained using Thompson Sampling, balancing exploration (learning which actions work), and exploitation (choosing the best-known action). Each customer instance was treated as a decision context, defined by their behavioral and economic profile. The agent selected between two actions—*Promotion* or *No Action*—and received rewards based on observed churn outcomes and CLV-weighted utility.

Fig. 1 presents early-stage results from the contextual bandit model for churn-sensitive promotions. For each customer, the model considered their predicted churn probability and CLV to choose an action—either *Promotion* or *No Action*. The reward was computed based on whether the customer stayed and their associated CLV, penalizing unnecessary promotions with a fixed cost when the customer still churned.

The summary reveals that "*No Action*" yielded a higher average reward (769.30) compared to "*Promotion*" (246.70). This indicates that in many cases, the model learned to avoid promotional spending on customers unlikely to churn or of low economic value. Initial simulations demonstrated that the bandit agent effectively learned to favor retention interventions for high-value or high-risk customers while avoiding unnecessary promotions for low-risk or low-value profiles.

The simulation was refined by incorporating actual churn probabilities predicted by Random Forest model, and promotion costs were adjusted to reflect realistic business conditions. The contextual bandit agent was re-run on the

updated dataset to evaluate the stability and effectiveness of its policy. Additionally, the evolution of the agent's decision-making behavior and its economic outcomes were visualized across different customer segments (Fig. 2):

- *Initial peak* (left side): A high promotion rate reflects random exploration when the agent is still uncertain about which actions are beneficial.

- *Gradual decline*: As the agent gathers evidence, it identifies that in many contexts, especially for low-value or

low-risk customers, promotions may not be economically justifiable.

- *Plateau and near-zero decisions* (right side): The agent stabilizes its policy, mostly avoiding promotions unless strongly indicated by customer context, thus minimizing wasted interventions.

	Churn_Prob	CLV	Action	Customer_Stays	Reward
0	0.866579	386.961240	Promotion	True	336.961240
1	0.873546	100.981938	No Action	False	0.000000
2	0.243370	164.802792	Promotion	True	114.802792
3	0.423268	180.403407	Promotion	False	-50.000000
4	0.343906	269.700620	Promotion	True	219.700620

Fig. 1. Early-stage results from the contextual bandit model

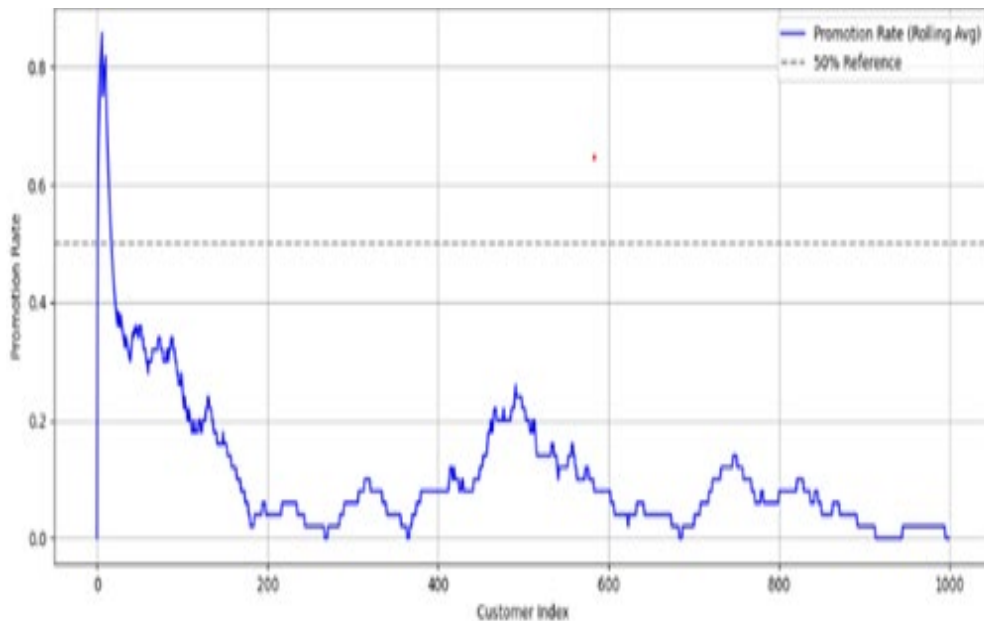


Fig. 2. Thompson Sampling: Evolution of promotion decision over time

This behavior aligns with optimal marketing strategy: prioritize retention resources only where the expected utility is positive, and learn adaptively to avoid over-investment in ineffective actions.

In the final stage, the Thompson Sampling contextual bandit was integrated with churn probabilities derived from a trained Random Forest classifier. The agent was applied to real, preprocessed customer data, which included economic (*CLV*) and behavioral (*Engagement Score*) features. Actions were selected based on expected utility, and rewards were computed from actual churn predictions and intervention outcomes.

Segmented performance analysis (Fig. 3) revealed that for **high CLV customers**, the “No Action” strategy consistently outperformed promotion, yielding average rewards exceeding 1900 compared to 780 from promotions. Among **low CLV customers**, both actions generated marginal returns, confirming their lower retention value. These results demonstrate the agent's learned preference to allocate promotions where economic return is highest, validating the use of contextual bandits as a utility-aware decision-making mechanism in customer retention systems.

The results (Fig. 4) clearly indicate that “No Action” consistently yields higher rewards across all churn risk segments. Specifically:

- In the *Low Risk* segment, the average reward for “No Action” exceeds 2000, while the reward for “Promotion” is significantly lower, around 450. This suggests that promoting to customers unlikely to churn often results in unnecessary costs with minimal benefit.

- In the *Medium Risk* group, the gap between “No Action” and “Promotion” narrows, but “No Action” still leads, indicating cautious targeting might be beneficial in this segment.

- In the *High Risk* segment, although both strategies yield lower rewards overall, “No Action” again results in a modest but superior outcome compared to “Promotion.”

These findings support the notion that indiscriminate promotional efforts, especially toward low-risk customers, can be economically inefficient. The reinforcement learning agent successfully learned to concentrate promotional efforts selectively, favoring high-utility outcomes and avoiding low-reward scenarios. The refinements enhanced the practical relevance of the model, demonstrating its readiness for integration into real-world churn management systems.

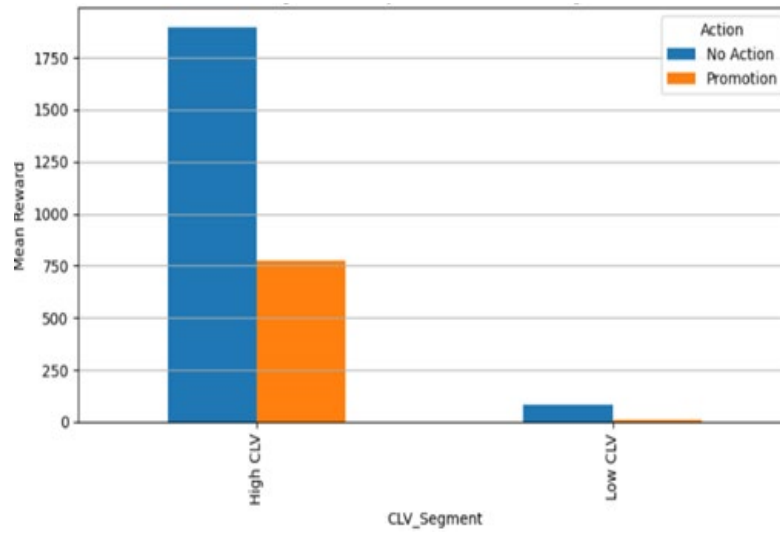


Fig. 3. Average reward by action within CLV segments

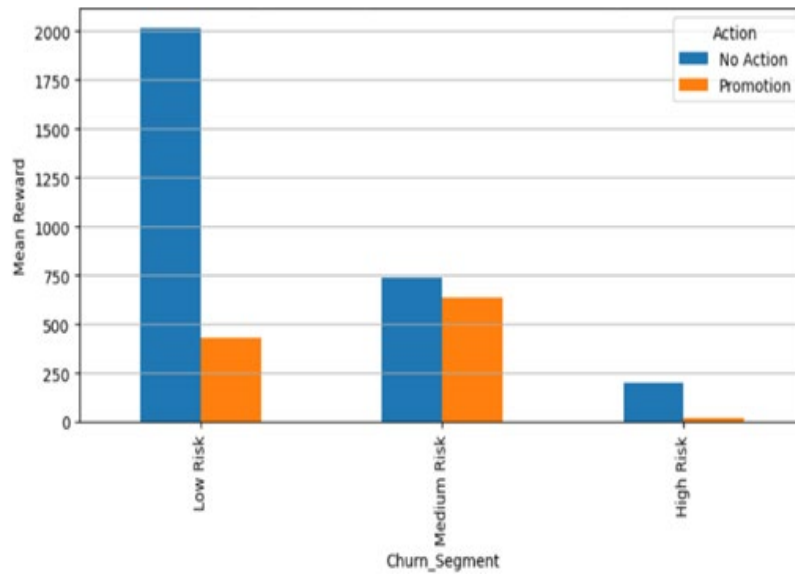


Fig. 4. Average reward by action within churn risk segments

IV. DISCUSSION

Applying reinforcement learning through a contextual bandit framework introduces a dynamic and utility-aware paradigm for managing churn-sensitive promotions. In contrast to static classification approaches, the agent continuously refines its decision policy by interacting with customer contexts, dynamically balancing exploration and exploitation. The use of Thompson Sampling facilitates probabilistic action selection under uncertainty, enabling the agent to initially explore a diverse action space and later converge to high-utility strategies. This adaptive behavior was empirically observed, with the promotion rate declining over time as the agent became more selective-favoring high-CLV customers where the expected economic gain surpassed intervention costs.

A key advantage of the proposed framework lies in its integration of Cumulative Prospect Theory (CPT) into the reward modeling process. By incorporating non-linear utility functions and probability distortion, the agent's decision-making process accounts for behavioral economics — specifically, customer risk perceptions and loss aversion. This allows the model to move beyond expected value

maximization, aligning churn management policies with realistic economic priorities.

However, the current simulation environment presents some limitations. The interaction model assumes static customer responses and fixed churn probabilities, without real-time feedback loops. As such, while the learning process is dynamic, the environment is not. To bridge this gap, future research could implement the agent within an online learning ecosystem, where customer behavior is continuously updated based on observed outcomes. This would enable the system to learn from live interactions, adapt to behavioral drift, and evolve strategies under non-stationary conditions.

Moreover, expanding the agent's action space to include multi-level promotions, discounts, or customer-specific communication channels would enhance its expressiveness. This extension could be supported by deep reinforcement learning techniques such as Deep Q-Networks (DQN) or Actor-Critic models, which are capable of approximating value functions over high-dimensional state-action spaces. Bayesian reinforcement learning, particularly Bayesian DQN or neural-linear contextual bandits, could further improve

exploration efficiency in uncertain environments by capturing posterior uncertainty over customer responses.

In addition, modeling temporal dependencies — such as recency, frequency, and customer lifecycle stages — via recurrent architectures or temporal attention mechanisms may improve the agent’s strategic foresight. Finally, incorporating seasonal trends and business cycles would allow the agent to adapt promotional policies to macro-level dynamics, ensuring robustness in real-world retail deployment.

V. CONCLUSION

This study introduced a reinforcement learning framework based on Contextual Bandits to enhance churn-sensitive promotional strategies in online retail. Unlike traditional classification models that predict churn risk in isolation, our approach dynamically adapts to customer behavior and economic context, optimizing promotional decisions over time using Thompson Sampling. The agent’s ability to prioritize high-value customers and allocate interventions selectively, guided by customer lifetime value, intervention cost, and behavioral utility, — demonstrates a meaningful shift toward economically rational churn management.

The integration of Contextual Bandits into churn frameworks marks a transition from reactive prediction to adaptive, reward-optimized decision-making. By explicitly incorporating the financial implications of retention strategies, this method not only improves predictive performance but also leads to smarter resource allocation, higher return on investment, and long-term profitability. Segment-level analyses confirmed that promotions are most effective in medium-risk segments and that indiscriminate actions in low-risk groups often yield negative returns.

From a business perspective, this model empowers organizations to move beyond one-size-fits-all retention strategies toward more personalized and value-sensitive interventions. This framework provides not only an adaptive modeling tool but also a strategic asset for marketing and retention teams aiming to maximize ROI in budget-constrained environments. It enables marketing teams to better justify promotional budgets, reduce wasted efforts, and focus on high-impact actions with measurable financial outcomes.

In practical terms, the proposed framework enables marketing teams to shift from broad retention campaigns to personalized, value-based interventions. This contributes to better budget allocation, improved ROI, and enhanced customer satisfaction. Moreover, the integration of economic utility functions allows for transparent, financially interpretable decision support — an essential factor in business adoption of Artificial Intelligence models.

Future research should focus on real-time deployment of the model, refinement through hierarchical or Bayesian bandits, and deeper integration of behavioral economic principles such as loss aversion and framing effects. Such advancements will enhance the robustness, interpretability, and business applicability of reinforcement learning systems in customer lifecycle management.

NOMENCLATURE

Symbol / Term	Definition
CLV	Customer Lifetime Value — the projected revenue attributed to the entire future relationship with a customer.
CPT	Cumulative Prospect Theory — a behavioral economic model accounting for risk sensitivity, loss aversion, and probability distortion.
V_{Gain}	Value function representing perceived gain under CPT.
V_{Loss}	Value function representing perceived loss under CPT.
P(churn)	Predicted probability that a customer will churn within a defined period.
Intervention Cost	The monetary cost associated with offering a promotion or incentive.
Expected Utility	Economic utility function combining P(churn), CLV, and intervention cost.
Contextual Bandit	A reinforcement learning algorithm that selects actions based on the current context to maximize cumulative reward.
Thompson Sampling	Bayesian method for action selection in bandit problems, balancing exploration and exploitation.
High-CLV / Low-CLV	Customers with above- or below-average lifetime value, respectively.

REFERENCES

- [1] Kahneman, Daniel, and Amos Tversky. “Prospect Theory: An Analysis of Decision under Risk.” *Econometrica*, vol. 47, no. 2, 1979, pp. 263–91. JSTOR, <https://doi.org/10.2307/1914185>. Accessed 23 Mar. 2025.
- [2] Verbraken, Thomas & Verbeke, Wouter & Baesens, Bart. (2012). A Novel Profit Maximizing Metric for Measuring Classification Performance of Customer Churn Prediction Models. *Knowledge and Data Engineering, IEEE Transactions on*. 25. 10.1109/TKDE.2012.50.
- [3] Bahnsen, A.C., Aouada, D. & Ottersten, B. A novel cost-sensitive framework for customer churn predictive modeling. *Decis. Anal.* 2, 5 (2015). <https://doi.org/10.1186/s40165-015-0014-6>.
- [4] Yuxiang Yuan, Chaofa Wang, Dong Cheng, Faming Zhang, Zhili Zhou, Faxin Cheng, Minimum conflict consensus models for group decision-making based on social network analysis considering non-cooperative behaviors, *Information Fusion*, Volume 99, 2023, 101855, ISSN 1566-2535, <https://doi.org/10.1016/j.inffus.2023.101855>. (<https://www.sciencedirect.com/science/article/pii/S1566253523001719>)
- [5] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction* — Cambridge, MA: MIT Press, 1998, 340 pp. ISBN 0-262-19398-1.
- [6] Li, Lihong, Wei Chu, John Langford, and Robert E. Schapire. "A contextual-bandit approach to personalized news article recommendation." In *Proceedings of the 19th international conference on World wide web*, pp. 661-670. 2010.
- [7] He, C., Ding, C.H.Q. A novel classification algorithm for customer churn prediction based on hybrid Ensemble-Fusion model. *Sci Rep* 14, 20179 (2024). <https://doi.org/10.1038/s41598-024-71168-x>
- [8] Liang, Zhiqing. (2023). Predict Customer Churn based on Machine Learning Algorithms. *Highlights in Business, Economics and Management*. 10. 270-275. <https://doi.org/10.54097/hbem.v10i.8051>.
- [9] Manzoor, Awais & Qureshi, M. & Kidney, Etain & Longo, Luca. (2024). A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners. *IEEE Access*. PP. 1-1. 10.1109/ACCESS.2024.3402092.
- [10] Payam Boozary, Sogand Sheykhan, Hamed GhorbanTanhaei, Cosimo Magazzino, Enhancing customer retention with machine learning: A comparative analysis of ensemble models for accurate churn prediction, *International Journal of Information Management Data Insights*, Volume 5, Issue 1, 2025, 100331